Google Cloud Run — Architecture Guide

Study reference for Google Cloud Professional Architect exam

What is Cloud Run?

Cloud Run is a fully managed compute platform that automatically scales stateless containers. It abstracts away all infrastructure management so you can focus on building great applications. Cloud Run is built on Knative, the open-source Kubernetes-based platform, and can run any container that listens for HTTP requests on a configurable port (default 8080).

Key Concepts

Services: The primary resource in Cloud Run. A service exposes a unique endpoint and automatically scales the underlying infrastructure to handle incoming requests.

Revisions: Every deployment creates an immutable revision. Traffic can be split between revisions for canary deployments and rollbacks.

Containers: Cloud Run executes your code packaged in a container image. Images must be stored in Artifact Registry or Container Registry.

Concurrency and Scaling

Each Cloud Run instance can handle multiple requests concurrently (default: 80, max: 1000). Cloud Run scales to zero when there are no incoming requests, which eliminates costs during idle periods. Minimum instances can be configured to keep warm instances available and avoid cold start latency.

Maximum instances cap the number of containers Cloud Run will start, which protects downstream services and controls costs. The default maximum is 100 instances per service.

Request Handling and Limits

Cloud Run has a maximum request timeout of 3600 seconds (1 hour). Requests that exceed this timeout are terminated. For long-running workloads, Cloud Run Jobs is the recommended alternative.

Memory limits range from 128 MiB to 32 GiB per instance. CPU is allocated only during request processing by default, but "CPU always allocated" mode keeps CPU available between requests for background tasks.

Startup CPU boost temporarily allocates additional CPU during instance startup to reduce cold start times.

Networking

Cloud Run services are accessible via HTTPS by default through an automatically provisioned URL. Custom domains can be mapped using Cloud Run domain mapping or a load balancer.

VPC Connector (Serverless VPC Access) allows Cloud Run to connect to resources in a VPC network, such as Cloud SQL instances or Memorystore. Traffic can be routed through the VPC for all outbound requests or only for private IP destinations.

Ingress controls determine which traffic can reach a service: All (public), Internal, or Internal and Cloud Load Balancing.

Identity and Access

Cloud Run services use a service account as their identity. By default, the Compute Engine default service account is used, but best practice is to create a dedicated service account with minimal required permissions.

Authentication can be required using IAM. Unauthenticated invocations must be explicitly allowed. The Cloud Run Invoker role (roles/run.invoker) grants permission to call a service.

Cloud Run Jobs

Cloud Run Jobs execute code that runs to completion rather than serving HTTP requests. Jobs are suitable for data processing pipelines, batch operations, and scheduled tasks.

A job consists of one or more tasks that can run in parallel. Each task runs one instance of the container. Tasks that fail can be retried up to a configurable maximum.

Pricing

Cloud Run pricing is based on CPU, memory, and requests:

• CPU: charged per vCPU-second during request processing (or always, if CPU always allocated)
• Memory: charged per GiB-second
• Requests: charged per million requests after free tier

The free tier includes 2 million requests, 360,000 vCPU-seconds, and 180,000 GiB-seconds per month. Minimum instance hours are charged at the CPU always-allocated rate even when idle.

Exam Key Points

1. Cloud Run scales to zero — eliminates costs when idle, introduces cold start latency.
2. Maximum concurrency default is 80 requests per instance; adjust based on workload memory needs.
3. Use minimum instances to guarantee low latency for latency-sensitive services.
4. VPC Connector required for private network resource access (Cloud SQL, Memorystore).
5. Cloud Run Jobs for batch workloads; Cloud Run Services for HTTP workloads.
6. Revisions are immutable — traffic splits enable safe progressive rollouts.